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Manufacturing method of split gate trench device

Abstract

A manufacturing method of a split gate trench device includes forming an epitaxial layer on a substrate, and forming a trench in the epitaxial layer, wherein the trench is divided into a first part and a second part above the first part. A shielding gate and a shielding oxide layer are then formed in the first part, wherein the shielding oxide layer is located between the shielding gate and the trench and exposes the second part. The second part is filled with an oxide, two grooves having a contour that is wide at the top and narrow at the bottom are then formed in the oxide, and a part of a sidewall of the trench is exposed. A gate oxide layer is formed on an exposed surface of the sidewall, and a first top gate and a second top gate are then formed in each of the two grooves.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 111138548, filed on Oct. 12, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a trench power device, and more particularly to a manufacturing method of a split gate trench device.

Description of Related Art

(3) Among power semiconductor devices, the power semiconductor device vertically disposed in the trench has become one of the focuses of development in all fields for being able to greatly increase the unit density.

(4) In the structure of the split gate trench power device, the gate structure in the power semiconductor device disposed in the trench is separated by an inter-poly oxide (IPO) layer and divided into two potentials. The upper gate is used to form the channel of the power device, and the lower gate is electrically coupled to the source potential with metal interconnection and is used to generate two-dimensional charge balance under the cut-off operation. In addition, in order to reduce the gate-source capacitance (C_{gs}), the upper gate may be changed into two separate structures, thereby improving the switching performance. However, such a structure reduces the gate (cross-sectional area), resulting in an increase in the gate resistance (R_g).

SUMMARY

(5) The disclosure provides a manufacturing method of a split gate trench device capable of manufacturing a gate having a structure that is wide at the top and narrow at the bottom, which can reduce the gate-source capacitance (C_{gs}) while improving the switching performance of the device.

(6) The manufacturing method of the split gate trench device of the disclosure includes forming an epitaxial layer on a substrate, and forming a trench in the epitaxial layer. The trench is divided into a first part and a second part, and the second part is located above the first part. Then, a shielding gate and a shielding oxide layer are formed in the first part of the trench. The shielding oxide layer is located between the shielding gate and the trench and exposes the second part. The second part of the trench is filled with an oxide, two grooves having a contour that is wide at top and narrow at bottom are then formed in the oxide, and a part of a sidewall of the trench is exposed. A gate oxide layer is formed on an exposed surface of the sidewall, and a first top gate and a second top gate are formed in each of the two grooves.

(7) In an embodiment of the disclosure, a method for forming the two grooves includes first forming a patterned mask layer on the oxide, exposing parts where the two grooves are to be formed, and then performing oxide dry etching.

(8) In an embodiment of the disclosure, a method for forming the two grooves includes first forming a patterned mask layer on the oxide, exposing parts where the two grooves are to be formed, then first performing oxide wet etching to form an upper groove having a curved surface, and then performing oxide dry etching to form a lower groove having a flat surface, so that the lower groove and the upper groove form the two grooves.

(9) In an embodiment of the disclosure, the patterned mask layer is formed above the shielding gate, and a projection of the patterned mask layer overlaps with a projection of the shielding gate on the substrate.

(10) In an embodiment of the disclosure, the substrate is divided into a device region and a peripheral region, and the step of forming the patterned mask layer includes fully covering the

peripheral region.

(11) In an embodiment of the disclosure, the method for forming the shielding gate and the shielding oxide layer includes conformally depositing the shielding oxide layer on an entire inner surface of the trench, filling the trench with a conductive material, then removing the conductive material outside the first part, exposing the shielding oxide layer located in the second part, and then removing the exposed shielding oxide layer.

(12) In an embodiment of the disclosure, the step of forming the trench includes respectively forming multiple trenches in the device region and the peripheral region.

(13) In an embodiment of the disclosure, before removing the conductive material outside the first part, the peripheral region may be covered with a photoresist layer.

(14) In an embodiment of the disclosure, before filling the second part of the trench with the oxide, the photoresist layer may also be removed.

(15) Based on the above, through first forming the two grooves having the contour that is wide at the top and narrow at the bottom in the inter-gate oxide layer, and then depositing to form the gate therein, the disclosure can increase the overall cross-sectional area of the gate. At the same time, since the gate is divided into two, the gate-source capacitance can be reduced, thereby reducing the switching loss.

(16) In order for the features and advantages of the disclosure to be more comprehensible, the following specific embodiments are described in detail in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic sectional view of a split gate trench device according to a first embodiment of the disclosure.

(2) FIG. 2 is a schematic sectional view of a split gate trench device according to a second embodiment of the disclosure.

(3) FIG. 3A to FIG. 3K are schematic sectional views of a manufacturing process of a split gate trench device according to a third embodiment of the disclosure.

(4) FIG. 4A and FIG. 4B are schematic sectional views of a manufacturing process of an alternative example of FIG. 3H according to the third embodiment.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

(5) The following disclosure provides many different implementations or examples for implementing different features of the disclosure. Of course, the embodiments are only examples and are not intended to limit the scope and application of the disclosure. Furthermore, the relative thicknesses and positions of various components, film layers, or regions may be reduced or exaggerated for clarity. In addition, similar or identical reference numerals are used in the drawings to denote similar or identical devices or features, and if there are reference numerals in the drawing that are identical to the previous drawing, the detailed description thereof will be omitted.

(6) FIG. 1 is a schematic sectional view of a split gate trench device according to a first embodiment of the disclosure.

(7) Please refer to FIG. 1. A split gate trench device **100** of the first embodiment basically includes a substrate **102**, an epitaxial layer **104** having a trench T, and a split gate structure SG. The trench T is divided into a first part t1 and a second part t2, and the first part t1 is located below the second part t2. The substrate **102** is a semiconductor substrate, such as a silicon substrate, and the epitaxial layer **104** is formed on the substrate **102**. In an embodiment, the substrate **102** is an N⁺ substrate, and the epitaxial layer **104** is an N⁻ epitaxial layer.

(8) The split gate structure SG is disposed in the trench T, and the split gate structure SG basically includes a shielding gate **106**, a first top gate **108a**, a second top gate **108b**, a shielding oxide layer

110, a gate oxide layer **112**, and an inter-gate oxide layer **114**. The shielding gate **106** is located inside (the first part **t1** of) the trench **T**, and the first top gate **108a** and the second top gate **108b** are respectively disposed above the shielding gate **106** and located in the second part **t2**. The shielding oxide layer **110** is located between the shielding gate **106** and the trench **T**, and the gate oxide layer **112** is located between the first top gate **108a** and the trench **T** and between the second top gate **108b** and the trench **T**. The materials of the shielding gate **106**, the first top gate **108a**, and the second top gate **108b** may be polysilicon or other suitable conductive materials.

(9) Please continue to refer to FIG. **1**. The inter-gate oxide layer **114** is located between the shielding gate **106**, the first top gate **108a**, and the second top gate **108b**. In detail, the inter-gate oxide layer **114** is an oxide structure obtained by an etching process and that is simultaneously between the shielding gate **106** and the first top gate **108a**, between the shielding gate **106** and the second top gate **108b**, and between the first top gate **108a** and the second top gate **108b**. Each of the first top gate **108a** and the second top gate **108b** has a shape that is wide at the top and narrow at the bottom, the wider part is away from the shielding gate **106**, and the narrower part is close to the shielding gate **106**. Since the shielding gate **106**, the first top gate **108a**, and the second top gate **108b** are usually polysilicon, the inter-gate oxide layer **114** is also referred to as an inter-poly oxide (IPO). The cross-sectional areas of the first top gate **108a** and the second top gate **108b** are greater than conventional polysilicon gates formed on two sides of the trench **T** by etching back, so a gate resistance R_g can be reduced.

(10) In an embodiment, a surface of the first top gate **108a** close to a sidewall of the trench **T** is a first flat surface **f1**, and a surface of the first top gate **108a** away from the sidewall of the trench **T** is a first inclined surface **i1**. A surface of the second top gate **108b** close to the sidewall of the trench **T** is a second flat surface **f2**, and a surface of the second top gate **108a** away from the sidewall of the trench **T** is a second inclined surface **i2**. Although the side wall of the first part **t1** of the trench **T** in FIG. **1** is shown as a flat surface, the disclosure is not limited thereto. In another embodiment, the entire sidewall of the trench **T** may be slightly inclined due to the manufacturing process, and the first flat surface **f1** and the second flat surface **f2** may also be slightly inclined flat surfaces, but such a change still enables the first top gate **108a** to have the shape that is wide at the top and narrow at the bottom, and the second top gate **108b** to have the shape that is wide at the top and narrow at the bottom. A bottom surface **b1** of the first top gate **108a** and a bottom surface **b2** of the second top gate **108b** are both flat surfaces, but the disclosure is not limited thereto.

(11) Please refer to FIG. **1** again. The first top gate **108a** and the second top gate **108b** are respectively disposed on the two sides in the trench **T**. Since FIG. **1** is a sectional view in a single direction, if the trench **T** in fact extends along another direction, the first top gate **108a** and the second top gate **108b** will also extend into long strips in the direction, and the shielding gate **106** will also extend into a long strip in the direction. In addition, if viewed from the floor plan, the first top gate **108a** will partially overlap with the shielding gate **106**, that is, a projection of the shielding gate **106** on the substrate **102** partially overlaps with a projection of the first top gate **108a** on the substrate **102**. Likewise, the projection of the shielding gate **106** on the substrate **102** partially overlaps with a projection of the second top gate **108b** on the substrate **102**. Such a configuration can prevent the gate resistance R_g from becoming too large.

(12) In FIG. **1**, the split gate trench device **100** may further include a body region **120** and a source region **122**. In the first embodiment, the body region **120** is formed in the epitaxial layer **104**, and a depth thereof is similar to a range of the first part **t1** of the trench **T**; and the source region **122** is formed in the body region **120** and is close to a top surface **104a** of the epitaxial layer **104**, wherein the body region **120** and the source region **122** have different conductivity, for example, the body region **120** is a P⁻ region, and the source region **122** is an N⁺ region. In addition, the source region **122** may be connected to interconnection (not shown) through a contact **126** formed in the dielectric layer **124**. The contact **126** may pass through the gate oxide layer **112**, the source region **122**, and a part of the body region **120** under the dielectric layer **124**, and another P⁺ region (not

shown) may be formed in a region on a bottom portion of the contact **126**. A drain electrode **130** may be disposed on a bottom portion **102b** of the substrate **102** to form a trench power device.

(13) FIG. **2** is a schematic sectional view of a split gate trench device according to a second embodiment of the disclosure, wherein the same or similar portions and components are represented by the same reference numerals as the first embodiment, and for the relevant content of the same or similar portions and components, reference may also be made to the content of the first embodiment, which will not be repeated.

(14) Please refer to FIG. **2**. The difference between a split gate trench device **200** of the second embodiment and the first embodiment lies in structures of a first top gate **202a** and a second top gate **202b** in the split gate structure SG. The first top gate **202a** and the second top gate **202b** also have the same shape that is wide at the top and narrow at the bottom, but a surface of the first top gate **202a** away from the sidewall of the trench T is composed of a first curved surface **c1** and a third flat surface **f3**, and a surface of the second top gate **202b** away from the side wall of the trench T is composed of a second curved surface **c2** and a fourth flat surface **f4**. The first curved surface **c1** is located above the third flat surface **f3**, and the second curved surface **c2** is located above the fourth flat surface **f4**. In an embodiment, an included angle between a tangent of a bottom portion of the first curved surface **c1** and the third flat surface **f3** is greater than 90 degrees, and an included angle between a tangent of a bottom portion of the second curved surface **c2** and the fourth flat surface **f4** is greater than 90 degrees.

(15) Since cross-sectional areas of the first top gate **202a** and the second top gate **202b** of the second embodiment are also greater than the conventional polysilicon gates formed on the two sides of the trench T by etching back, the gate resistance R_g can be reduced. Moreover, the projection of the shielding gate **106** on the substrate **102** partially overlaps with a projection of the first top gate **202a** and the second top gate **202b** on the substrate **102**, so the gate resistance R_g will not become too large.

(16) FIG. **3A** to FIG. **3K** are schematic sectional views of a manufacturing process of a split gate trench device according to a third embodiment of the disclosure.

(17) In FIG. **3A**, a substrate **300** is divided into a device region **30a** and a peripheral region **30b**. The third embodiment mainly focuses on the manufacturing of a split gate trench device in the device region **30a**, and can simultaneously manufacture a trench gate structure in the peripheral region **30b**. However, the disclosure is not limited thereto. In another embodiment, the trench gate structure in the peripheral region **30b** may be manufactured by adopting different manners. Please refer to FIG. **3A** first. An epitaxial layer **302** is formed on the substrate **300**, the trench T is then formed in the epitaxial layer **302**, and the trench T is divided into the first part **t1** and the second part **t2**, wherein the second part **t2** is located above the first part **t1**. The method for forming the trench T is to, for example, first form an oxide layer and a hard mask (not shown) on a surface **302a** of the epitaxial layer **302**, then expose the surface **302a** of the epitaxial layer **302** to form the trench T using a photolithography etching process, and then dry etch (anisotropic etch) the epitaxial layer **302** using the hard mask as an etching mask to form the trench T.

(18) Then, please refer to FIG. **3B**. In order to form a shielding gate and a shielding oxide layer, a shielding oxide layer **304** may be first conformally deposited on an entire inner surface of the trench T. At the same time, the shielding oxide layer **304** is also deposited on the surface **302a** of the epitaxial layer **302**.

(19) Then, please refer to FIG. **3C**. The trench T is filled with a conductive material **306**. The formation of the conductive material **306** is to, for example, first fully deposit polysilicon, and then fully etch back the polysilicon to remove polysilicon outside the trench T.

(20) Next, please refer to FIG. **3D**. The peripheral region **30b** is covered with a photoresist layer **PR1** to ensure that the conductive material **306** of the peripheral region **30b** is not affected by subsequent processes. The conductive material **306** outside the first part **t1** of the trench T is removed, and the shielding oxide layer **304** located in the second part **t2** is exposed.

(21) Then, please refer to FIG. 3E. The shielding oxide layer **304** exposed in FIG. 3D is removed to form a shielding gate **308** and a shielding oxide layer **304'** in the first part **t1** of the trench **T**, wherein the shielding oxide layer **304'** is located between the shielding gate **308** and the trench **T** and exposes the second part **t2**.

(22) After that, please refer to FIG. 3F. After removing the photoresist layer **PR1**, the second part **t2** of the trench **T** is filled with an oxide **Ox**, wherein the method for forming the oxide **Ox** is to, for example, perform full deposition using high-density plasma chemical vapor deposition (HDP-CVD).

(23) Then, please refer to FIG. 3G. If the oxide **Ox** fully covers the device region **30a** and the peripheral region **30b** as shown in FIG. 3F, a chemical mechanical planarization process may be first performed on the oxide **Ox**, and etch back is then performed to remove the oxide **Ox** outside the trench **T**. Next, a patterned mask layer **310** is formed on the oxide **Ox**, and parts where two grooves to be formed are exposed. In an embodiment, the patterned mask layer **310** is formed above the shielding gate **308** and a projection thereof overlaps with a projection of the shielding gate **308** on the substrate **300**. In addition, the patterned mask layer **310** may also fully cover the peripheral region **30b** to ensure that the structure of the peripheral region **30b** is not affected by subsequent processes.

(24) Then, please refer to FIG. 3H. Oxide dry etching is performed to form two grooves **312a** and **312b** having a contour that is wide at the top and narrow at the bottom in the oxide **Ox**, and expose a part of the sidewall of the trench **T**. The oxide dry etching forms a groove contour having an inclined surface using the patterned mask layer **310** as an etching mask, while applying anisotropic etching characteristics of dry etching to etch down.

(25) Next, please refer to FIG. 3I. All of the patterned mask layer **310** in FIG. 3H is first removed, a gate oxide layer **314** is then formed on a surface of the sidewall **T** exposed in FIG. 3H, and the gate oxide layer **314** may be simultaneously formed on the surface **302a** of the epitaxial layer **302** and a surface of the conductive material **306** in the peripheral region **30b**. Then, an entire layer of the conductive material **316**, such as polysilicon, is deposited on the epitaxial layer **302**.

(26) Then, please refer to FIG. 3J. After a planarization step, a first top gate **318a** and a second top gate **318b** may be formed in each of the two grooves **312a** and **312b** in FIG. 3H to complete a split gate structure including the shielding gate **308**, the first top gate **318a**, the second top gate **318b**, the shielding oxide layer **304'**, the gate oxide layer **314**, and the inter-gate oxide layer (the oxide **Ox**). After that, a body region **320** and a source region **322** may be formed in the epitaxial layer **302**, and the forming method is to, for example, first implant p-type dopants from the surface **302a** by ion implantation to form the P⁻ body region **320**, and then implant n-type dopants from the surface **302a** by ion implantation to form the N⁺ source region **322**. At the same time, the body region **320** and the source region **322** may also be formed in the peripheral region **30b**.

(27) After that, please refer to FIG. 3K. An entire layer of a dielectric layer **324** may be first formed on the structure of FIG. 3J, and a contact **326** is then formed therein and passes through the gate oxide layer **314**, the source region **322**, and a part of the body region **320** under the dielectric layer **324**, wherein the method for forming the contact **326** is to, for example, form a patterned photoresist layer (not shown) on the dielectric layer **324** after forming the dielectric layer **324** to expose a part where the contact **326** is to be formed, then etch to form an opening **326a** passing through the dielectric layer **324**, the gate oxide layer **314**, the source region **322**, and a part of the body region **320**, and then fill the opening **326a** with a conductive material to form the contact **326**. In addition, after forming the opening **326a** and before filling the conductive material, the p-type dopants may be implanted from the opening **326a** by ion implantation to pre-form a P⁺ region (not shown) in a region on the bottom portion of the contact **326**. After that, source metal layers **328** connecting the contacts **326** of the device region **30a** and the peripheral region **30b** are formed. The conductive material **306** in the peripheral region **30b** may be electrically connected to the source region **322** of the device region **30a** through the source metal layer **328**, and in the embodiment, the

shielding gate **308** and the conductive material **306** in the peripheral region **30b** are of the same structure and are connected to each other.

(28) In addition to the above process of the third embodiment, FIG. **3H** may also use the following steps instead.

(29) First, please refer to FIG. **4A**. After forming the patterned mask layer **310** on the oxide **Ox** and exposing parts where grooves are to be formed, oxide wet etching may be first performed to form an upper groove **400a** having a curved surface, wherein the oxide wet etching here is to, for example, etch using a buffered hydrofluoric acid (HF) or a diluted HF as an etchant. Since wet etching is isotropic etching, a range of the upper groove **400a** will expand to below the patterned mask layer **310**, but there will be a predetermined distance between the two upper grooves **400a**.

(30) Next, please refer to FIG. **4B**, oxide dry etching is performed to form a lower groove **400b** having a flat surface, wherein the oxide dry etching also uses the patterned mask layer **310** as an etching mask, so a cross-sectional shape of the lower groove **400b** does not change much. A groove composed of the lower groove **400b** and the upper groove **400a** that is wide at the top and narrow at the bottom is obtained, so following the steps of FIG. **3I** to FIG. **3J**, the first top gate **202a** and the second top gate **202b** as shown in FIG. **2** may be obtained.

(31) In summary, the method of the disclosure is to first form the two grooves having the contour that is wide at the top and narrow at the bottom in the inter-gate oxide layer, and then deposit polysilicon therein to form the gate that is wide at the top and narrow at the bottom, so the overall cross-sectional area of the gate can be increased, thereby reducing the gate resistance. At the same time, since the top gate of the split gate trench device is divided into two, the gate-source capacitance can be reduced, thereby reducing the switching loss.

(32) Although the disclosure has been disclosed in the above embodiments, the embodiments are not intended to limit the disclosure. Persons skilled in the art may make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the appended claims.

Claims

1. A manufacturing method of a split gate trench device, comprising: forming an epitaxial layer on a substrate; forming a trench in the epitaxial layer, wherein the trench is divided into a first part and a second part, and the second part is located above the first part; forming a shielding gate and a shielding oxide layer in the first part of the trench, wherein the shielding oxide layer is located between the shielding gate and the trench and exposes the second part; filling the second part of the trench with an oxide; forming two grooves having a contour that is wide at top and narrow at bottom in the oxide, and exposing a part of a sidewall of the trench; forming a gate oxide layer on an exposed surface of the sidewall; and forming a first top gate and a second top gate in each of the two grooves.
2. The manufacturing method of the split gate trench device according to claim 1, wherein a method for forming the two grooves comprises: forming a patterned mask layer on the oxide, and exposing parts where the two grooves are to be formed; and performing oxide dry etching.
3. The manufacturing method of the split gate trench device according to claim 2, wherein the patterned mask layer is formed above the shielding gate, and a projection of the patterned mask layer on the substrate overlaps with a projection of the shielding gate on the substrate.
4. The manufacturing method of the split gate trench device according to claim 2, wherein the substrate is divided into a device region and a peripheral region, and the step of forming the patterned mask layer comprises fully covering the peripheral region.
5. The manufacturing method of the split gate trench device according to claim 1, wherein a method for forming the two grooves comprises: forming a patterned mask layer on the oxide, and exposing parts where the two grooves are to be formed; performing oxide wet etching to form an

upper groove having a curved surface; and performing oxide dry etching to form a lower groove having a flat surface, wherein the lower groove and the upper groove form the two grooves.

6. The manufacturing method of the split gate trench device according to claim 5, wherein the patterned mask layer is formed on the shielding gate, and a projection of the patterned mask layer on the substrate overlaps with a projection of the shielding gate on the substrate.

7. The manufacturing method of the split gate trench device according to claim 5, wherein the substrate is divided into a device region and a peripheral region, and the step of forming the patterned mask layer comprises fully covering the peripheral region.

8. The manufacturing method of the split gate trench device according to claim 1, wherein a method for forming the shielding gate and the shielding oxide layer comprises: conformally depositing a shielding oxide layer on an entire inner surface of the trench; filling the trench with a conductive material; removing the conductive material outside the first part, and exposing the shielding oxide layer located in the second part; and removing the exposed shielding oxide layer.

9. The manufacturing method of the split gate trench device according to claim 8, wherein the substrate is divided into a device region and a peripheral region, and the step of forming the trench comprises respectively forming a plurality of trenches in the device region and the peripheral region.

10. The manufacturing method of the split gate trench device according to claim 9, wherein before removing the conductive material outside the first part, the manufacturing method further comprises covering the peripheral region with a photoresist layer.

11. The manufacturing method of the split gate trench device according to claim 10, wherein before filling the second part of the trench with the oxide, the manufacturing method further comprises removing the photoresist layer.
